

# Separating Chaos From Noise

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# 1 Introduction

Chaos and noise are often interpreted as synonymous, however their difference is integral to the motivation behind this project. When a system is referred to as chaotic, its inherent unpredictability, sensitivity, and nonlinearity become its main properties. Furthermore, a noisy model emphasizes its short-term randomness and stochastic behaviour. When a system is both chaotic and noisy, it is interesting to examine its statistical properties and their origin. This project explores whether a statistical inference is a result of the fundamental chaos of the system or the stochastic noise imposed on it.

Named after mathematician and meteorologist Edward Lorenz who first studied the model in 1963, the Lorenz system(5) is a deterministic nonlinear system that became a fundamental model of chaos. Chaos, although seemingly unpredictable, is contained within certain boundaries and abides by statistical conditions like the fixed points. When randomness in the form of Gaussian white noise, which is time independent and normally distributed, is introduced into the the Lorenz system, these statistical conditions that keep the chaos contained experience changes, and these changes are studied in this paper. This enables a deeper understanding of this interaction between chaos and noise, which can then be applied to more physically relevant models.

A combination of chaos and noise is observed everyday, from climate and health data to market trends and economic fluctuations. In these models, chaos is inherent to the data being studied, like neural activity or heart rate variability in health data to inflation and interest rate dynamics and global trade network models. The noise in this context can be attributed to genetic variability and environmental factors to exogenous shocks and market fluctuations. A better understanding of this interaction of chaos and noise leads to an improved understanding of real life patterns. This is further used to develop predictions that can be crucial for saving lives and forming better investment strategies. The methods introduced in this paper can be pivotal to understanding and optimising these models, which can enable better inference and forecasting.

Noise is introduced to the Lorenz system by being discretised using the Euler Maruyama method, which enables easy computation of the model to derive valuable inferences. The Probability Density Functions of this process are then examined to observe its behaviour in the phase space. The Power Law analysis determines the extent of scale-free behaviour or autocorrelation in the system. These qualities change with noise, and this insight into the effect of stochastic noise onto these systems achieves the ultimate aim.

## 1.1 Ethical Considerations

The Lorenz system is a heavily truncated physical model, and the results derived in this project are not robust enough for any physical inference and are only meant for mathematical purposes. The methods used introduce errors into the system, that may accumulate to be substantial, hence making the final conclusions unsuitable for physical interpretation. The chaotic nature of the model means that, even with the same initial conditions and parameters, the results generated every time the code is run may vary. Therefore, outcomes should be interpreted carefully, as they may not always be fully consistent and reliable.

## 2 The Lorenz System

In his paper 'Deterministic Nonperiodic Flow' (5), Edward Lorenz modelled atmospheric convection using three nonlinear differential equations, known as the Lorenz system. Lorenz studied the movement of heat in the atmosphere via convection. Since solving the full set of equations was too complex, he simplified the atmospheric fluid dynamics equations into a deterministic model that takes the following form:

$$\begin{aligned}\frac{dx}{dt} &= \sigma(y - x) \\ \frac{dy}{dt} &= x(\rho - z) - y \\ \frac{dz}{dt} &= xy - \beta z\end{aligned}\tag{2.1}$$

Where:

$x$  = rate of convection.

$y$  = temperature difference between the ascending and descending currents.

$z$  = vertical temperature difference in the atmosphere

$\sigma$  = Prandtl number, representing the ratio of fluid viscosity to thermal conductivity

$\rho$  = Rayleigh number, representing the difference driving the convection

$\beta$  = ratio of the width and height of the convection layer

Lorenz derives this model from Barry Saltzman's paper titled "Finite Amplitude Free Convection as an Initial Value Problem—I" in 1962. In his original paper (5), Lorenz used  $\sigma = 10$  and  $\beta = \frac{8}{3}$ . The critical Rayleigh number for the instability of steady convection is  $\rho = \frac{470}{90} = 24.74$ , and Lorenz uses a slightly supercritical value of  $\rho = 28$ . These constant values produce a trajectory in phase space called the Lorenz Attractor because of its butterfly-like structure.

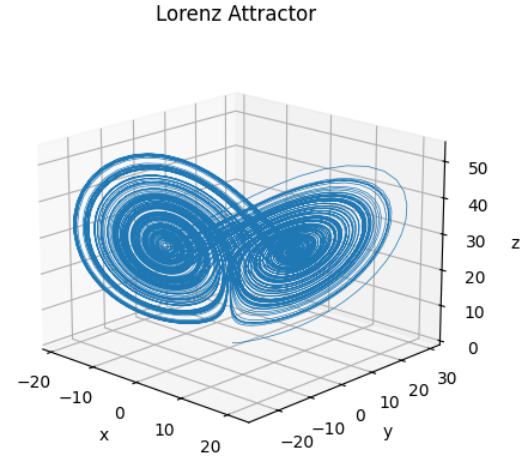


Figure 1: This is the Lorenz Attractor for the original values of  $\sigma$ ,  $\rho$  and  $\beta$ .

### 2.1 Linear Stability Analysis

The fixed points of the attractor exist when all the equations in 2.1 are 0. Solving these equations reveals that the fixed points are

$$\begin{aligned}&(0, 0, 0) \\ &(\sqrt{\beta(\rho - 1)}, \sqrt{\beta(\rho - 1)}, \rho - 1) \\ &(-\sqrt{\beta(\rho - 1)}, -\sqrt{\beta(\rho - 1)}, \rho - 1)\end{aligned}\tag{2.2}$$

These fixed points plotted on an attractor and projected onto a 2 dimensional X-Z and Y-Z space are plotted in Figure 2

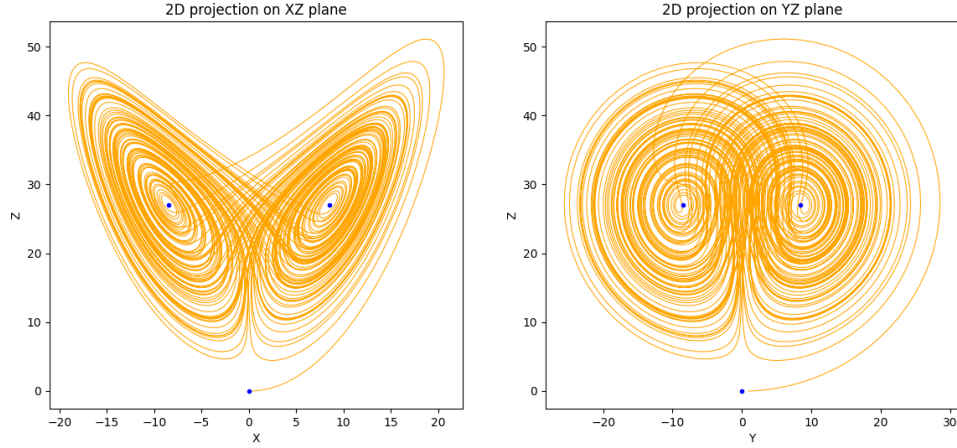


Figure 2: The blue dots are the fixed points of the attractor.

## 2.2 Sensitivity to Initial Conditions

The Lorenz system exhibits extreme sensitivity to initial conditions and this is an integral characteristic of a chaotic system.

Figure 3 shows this by creating two simulations of the Lorenz system with the initial conditions  $(x_1, y_1, z_1) = (1, 1, 1)$  and  $(x_2, y_2, z_2) = (1, 1, 1.0000001)$ . *Subplot1* in Figure 3 plots values of  $x_1$  for the first 6000 points in the Lorenz simulation (1). Although  $z_2$  is closely approximated by  $z_1$ , this small change can be observed in the difference  $x_1 - x_2$  plotted in *subplot2* in Figure 3.

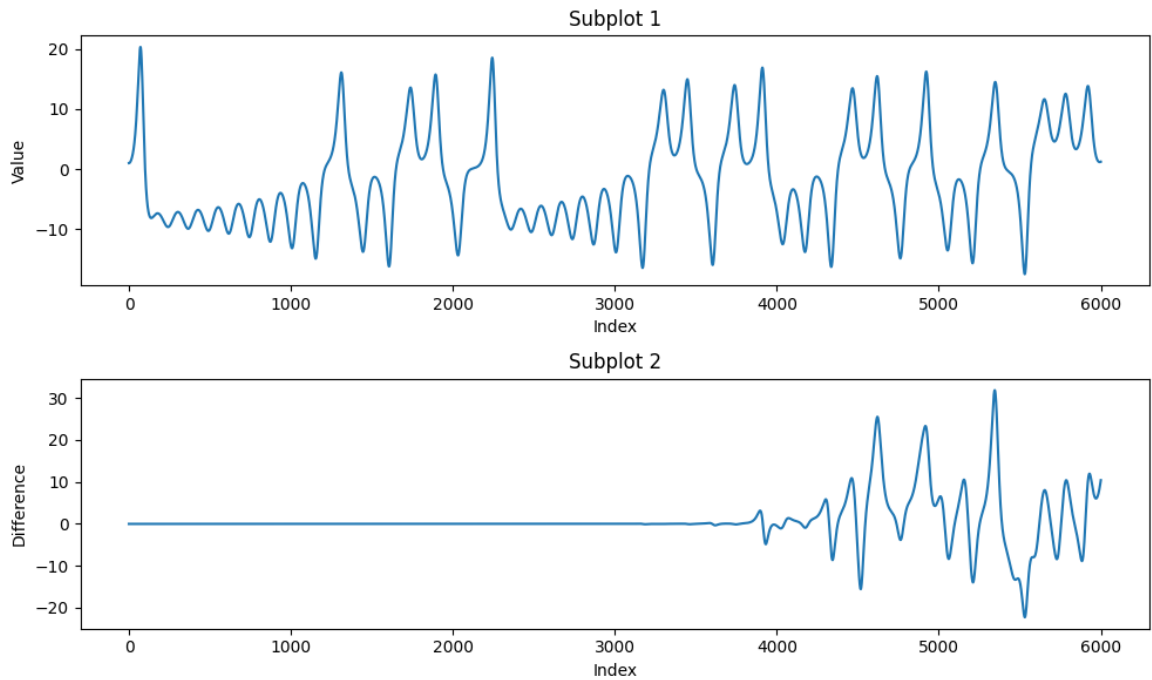


Figure 3: Sensitivity to Initial Conditions

Both simulations plotted in the 3 dimensional space are shown in Figure 4

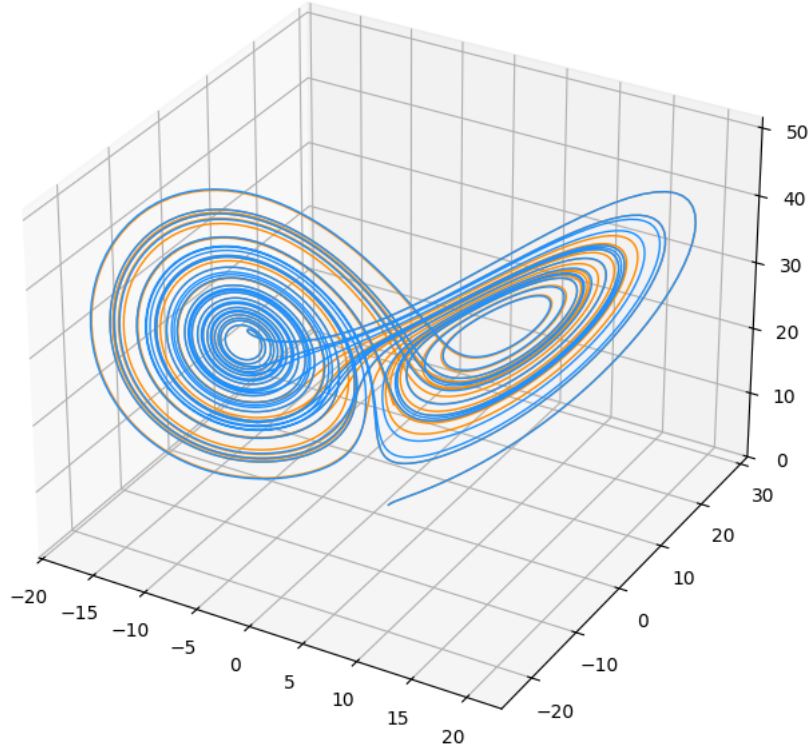


Figure 4: The two attractors are simulations of the Lorenz system with minutely different initial conditions.

### 3 Langevin Equation

The Langevin equation is a stochastic differential equation(SDE) introduced by Paul Langevin in 1908. He derived this from Newton's second law of motion that states:

$$m \frac{dv}{dt} = F \quad (3.1)$$

Where:

$F$  is the force acting on the Brownian particle from the surrounding fluid.

Langevin suggested that force can be expressed in two parts: friction force or drag,  $F_A$ , which represents the friction experienced by the particle and is proportional to its velocity. The second part of Newton's force,  $F_B$ , represents the random force, also known as Gaussian White noise, which is stochastic in nature. This term is  $\delta$  - correlated, which means it has memoryless-ness property(8). This equation describes the time evolution of the position of a Brownian particle and takes the form:

$$\begin{aligned} m \frac{dv}{dt} &= F_A + F_B \\ &= -m\gamma v + F_B \end{aligned} \quad (3.2)$$

Where:

$m$  = mass of the particle

$v$  = velocity

$\gamma$  = drag coefficient

$F_B$  = Stochastic noise

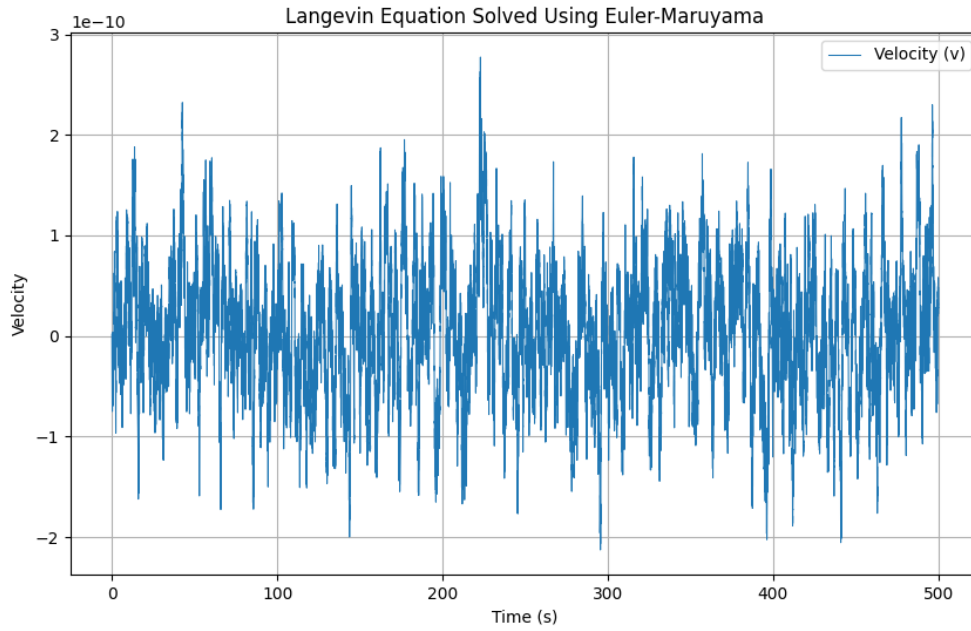


Figure 5: Plot of Brownian Motion against Time

Figure 5 shows a plot of the velocity of a particle in Brownian motion against time using the Euler-Maruyama method. It is also worth noting that since every next displacement of the molecule is dependant only on its current position, the Langevin equation qualifies as a Markov Process, or can be described as a random walk. The correlation function for the Langevin system decays exponentially.

### 3.1 Errors - Global and Local

Since Figure 5 is generated via code and uses the Taylor expansion/Euler Maruyama Method, it is susceptible to Local errors caused by the use of insufficient terms of the Taylor expansion. This leads to a truncation error from approximating derivatives via the Taylor Expansion. This occurs at every step and hence reflects into every next point calculated in the code.

A Global error is observed when these errors accumulate over multiple timesteps. This results into the difference between the exact and the numerical solutions.

### 3.2 Ornstein-Uhlenbeck process

The Ornstein-Uhlenbeck process is a form of the Langevin equation that exhibits linear drift and constant diffusion. It takes the form:

$$m \frac{dv}{dt} = -\gamma v + \sqrt{2\gamma kT} \xi(t) \quad (3.3)$$

Where:

$m$  = mass of the particle

$v$  = velocity

$\gamma$  = drag coefficient

$kT$  = Boltzmann constant, also known as the temperature constant

$\xi(t)$  = Gaussian white noise

Since this process observes mean reversion, and is classified as stationary, meaning that its statistical properties do not change over time. This is an excellent equation to validate the Linear Stability Analysis and the Detrended Fluctuation Analysis, before it is implemented onto the Lorenz system. For simplicity, this version of the Langevin system is hereby referred to as the Langevin system.

## 4 Probability Density Functions

The steady state solution of a nonlinear system gives the long term probability density function (PDF), which enables a deeper understanding of the system through visualizations and inference of important statistical properties like the peaks of this diagram indicating the fixed points of the system.

### 4.1 Probability Density Function for Langevin equation

The PDF of a chaotic system is approximated by plotting a normalized histogram. Figure 6 plots the PDF for the velocities of the Langevin equation. The PDF is better approximated with more steps.

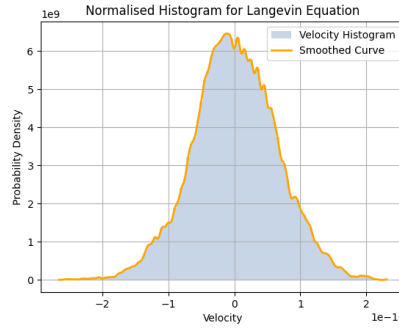


Figure 6: Normalised histograms approximating the probability density function for the Langevin equation

### 4.2 Fokker-Planck Equation

The Fokker-Planck equation is a robust method of obtaining the analytical steady state solution for a stochastic differential equation like the Langevin equation (6). The green line in Figure 7 reproduces the Fokker-Planck equation alongside the normalised histogram to show this approximation.

The steady-state solution exists when  $\frac{\partial P(v,t)}{\partial t} = 0$ .

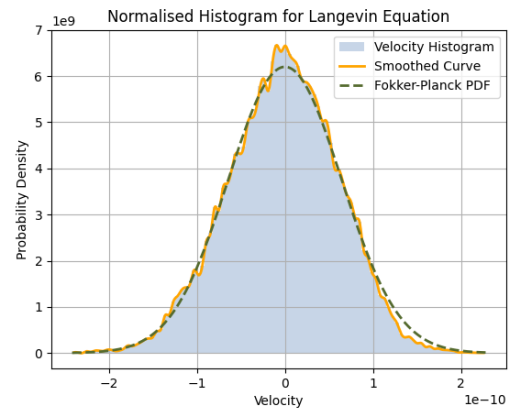


Figure 7: Fokker-Planck PDF of Langevin Equation

Therefore, it is validated that normalised histograms are a close approximation of the Fokker Planck equation, which is the steady state solution to the Langevin equation.

### 4.3 Lorenz PDFs

Figure 8 plots the normalized histograms for the  $x$ ,  $y$  and  $z$  values of Equation 2.1, which can be claimed to approximate the PDF.

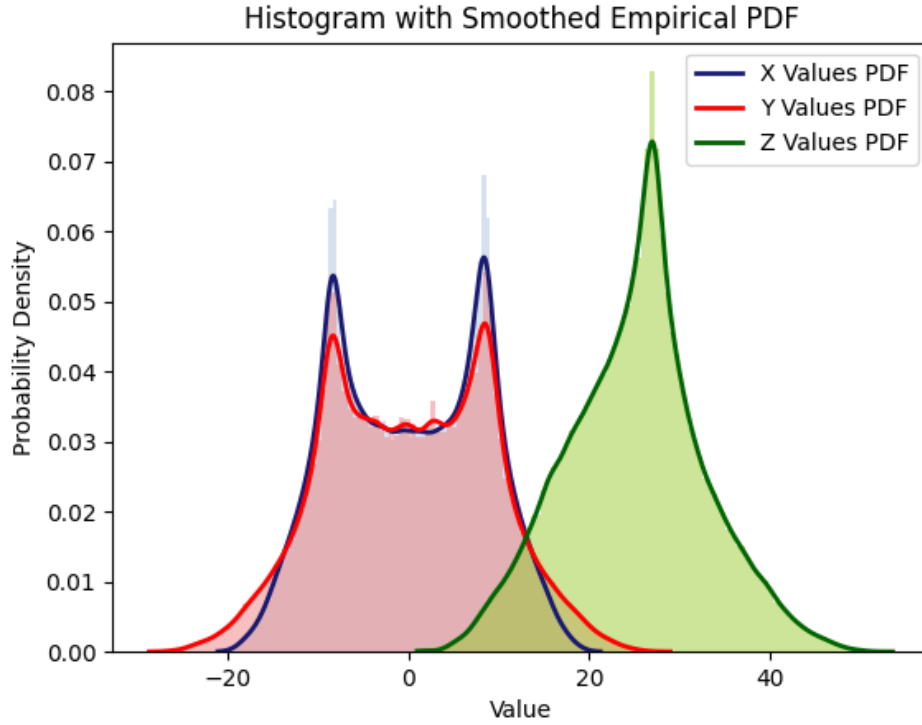


Figure 8: Normalised histograms of the Lorenz system

It is observed that the PDFs for the  $x$  and  $y$  axes are bimodal distributions. This shows the location of the two non trivial fixed points of the attractor. On the other hand, the distribution for the  $z$  axis is unimodal because the two non trivial fixed points take the same value on the  $z$  plane. These PDFs aid the understanding of the Lorenz system, and the changes that will be observed with the introduction of noise will be integral to the paper's aim.

## 5 Noise and Power Spectral Density

Adding noise to a process means introducing a stochastic component that will cause deviations from its original trajectory. Figure 9 shows this change in a simple linear equation, and the 'Noisy Line' plot is mathematically referred to as a *Random Walk*.



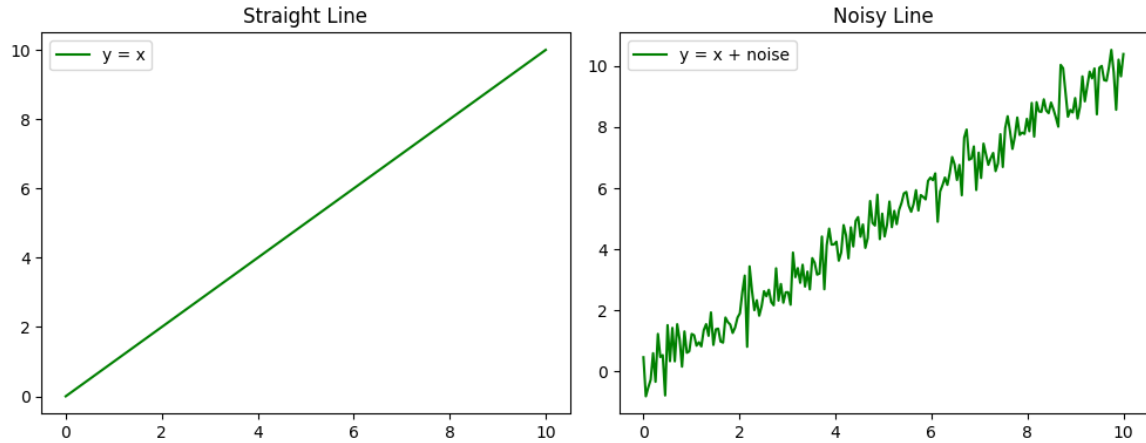


Figure 9: A linear plot with added noise

## 5.1 Time Step Independence

When noise is introduced to a system, it is important to ensure its time step independence, so that the magnitude of noise added across a said period of time is uniform and does not depend on the length of timestep. This means that the cumulative noise added to a section on the system is independent of how big the time step is.

```
for i in range(200000):
    dx = sigma*(yval - xval)
    dy = xval*(rho - zval) - yval
    dz = xval*yval - beta*zval

    xval = xval + dt*dx + magnitude * np.sqrt(dt) * np.random.
        normal()
    yval = yval + dt*dy + magnitude * np.sqrt(dt) * np.random.
        normal()
    zval = zval + dt*dz + magnitude * np.sqrt(dt) * np.random.
        normal()
    coord.append((xval, yval, zval))
```

The code above adds noise in three components: *magnitude*,  $\sqrt{\text{timestep}}$  and a *random number* from the standard normal distribution. While *magnitude* and *random number* introduce the noise of the desired magnitude, the  $\sqrt{\text{timestep}}$  term ensures that this noise is independent of the timestep defined to implement the Euler-Maruyama method.

## 5.2 Power Spectral Density

The Power Spectral Density(PSD) illustrates a signal's distribution of power at different frequencies. The spectral exponent, or the slope obtained in the log-log plot is an indication of the energy per unit frequency.

The main aim of this project is to introduce noise to a deterministic nonlinear system like the Lorenz system and to analyse the effects. Noise is Gaussian when it is truly random and does not depend on time or the current signal. This randomness follows a Gaussian or

normal distribution, which has a bell-shaped probability distribution.

### 5.3 Coloured Noise

Stochastic noise is classified into several *colours*, each of which is likely to affect a chaotic system differently. These colours are classified by the power spectral density(PSD) of noise, which follows the following form:

$$PSD \propto \frac{1}{f^\alpha} \quad (5.1)$$

Where  $f$  = frequency and  $\alpha \geq 0$ .  $\alpha$  is the spectral exponent and corresponds to a different colour of noise (13). Equation 5.1 is also referred to as *Power Law*, which determines the extent of the scale free behaviour exhibited by the process(12).

#### 5.3.1 White Noise

Noise is called white when for Equation 5.1,  $\alpha = 0$ , which also indicates its slope on the log-log plot. This results in constant power spectral density, which means it has equal power at any frequency. This is visually represented on a log-log plot in Figure 10.

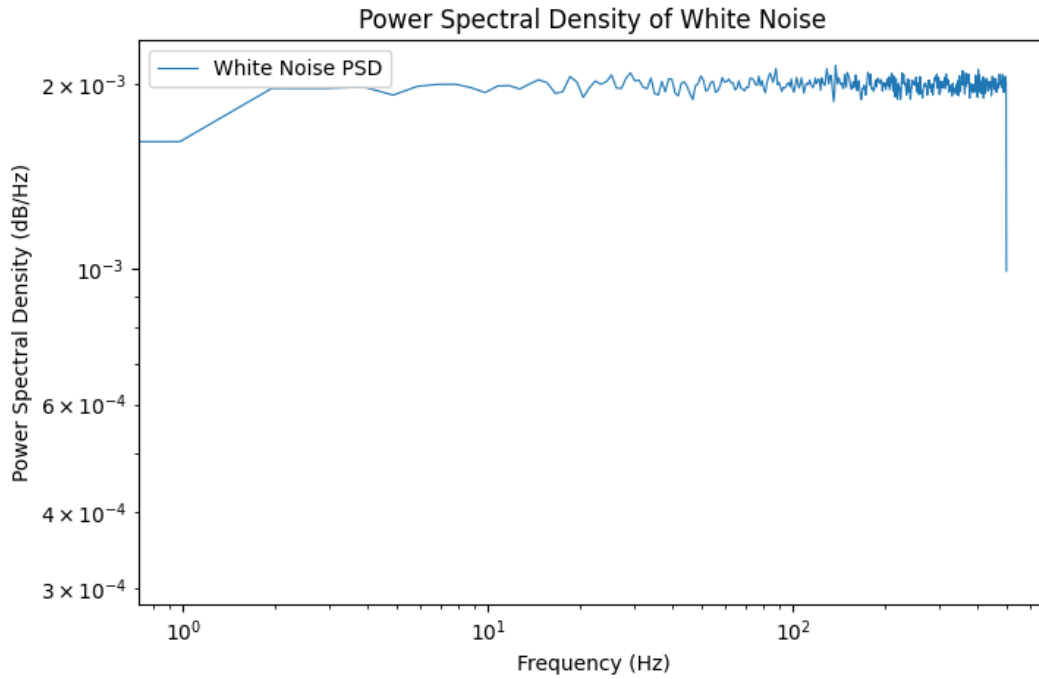


Figure 10: Power Spectral Density of White Noise: The flat spectrum indicates that power is evenly distributed across all frequencies, a characteristic feature of white noise.

White noise is time independent and has a  $\delta$  correlated autocorrelation function.

#### 5.3.2 Red or Brown Noise

Noise is called red or brown when for Equation 5.1,  $\alpha = 2$ , which also indicates its slope on the log-log plot. This means the power spectral density decays at a rate of  $\frac{1}{f^2}$ . This is created

by Brownian motion and has more energy at lower frequency. It is generated by integrating white noise and is visually represented on a log-log plot in Figure 11

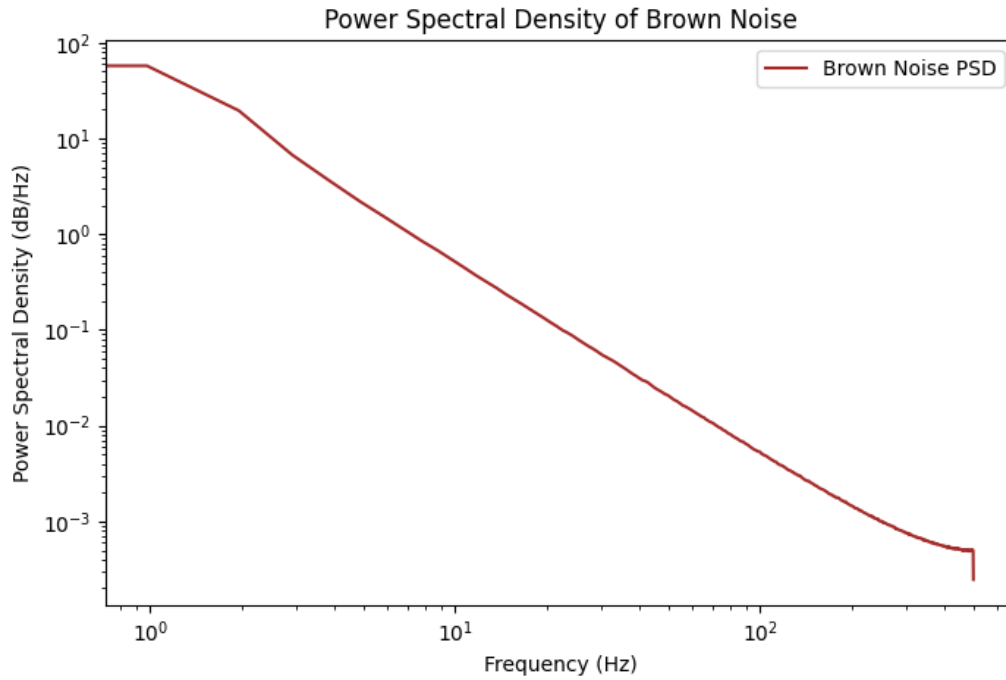


Figure 11: Power Spectral Density of White Noise: The flat spectrum indicates that power is evenly distributed across all frequencies, a characteristic feature of white noise.

## 5.4 Noise in the Lorenz System

This section observes the effect of adding Gaussian white noise to the Lorenz system.

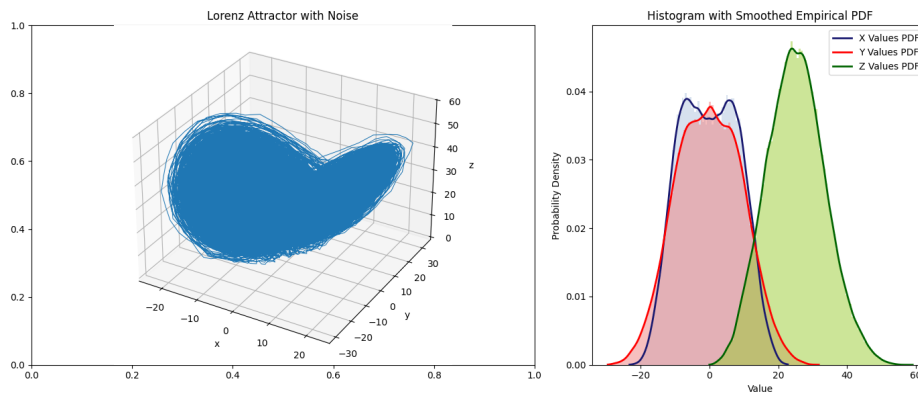


Figure 12: Normalised histogram of the Lorenz system with added noise of magnitude 5

Figure 12 shows that the introduction of Gaussian white noise results in data points positioned comparatively more uniformly throughout the system domain, unlike its naturally bimodal distribution on  $x$  and  $y$ . There is still significant overlap, and the change in the PDF of  $z$  is minimal. The attractor is noisy, but still maintains an essence of its winged structure in the three-dimensional space.

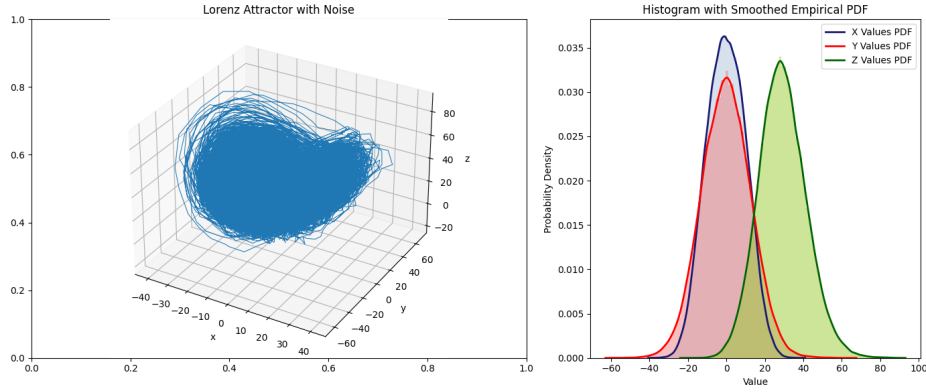


Figure 13: Normalised histogram of the Lorenz system with added noise of magnitude 20

Interestingly, Figure 13, where the noise is of magnitude 20, exhibits a unimodal PDF for all  $x$ ,  $y$  and  $z$ . Here, the noise has a much larger impact on both the PDFs and the attractor itself. The attractor loses the prominence its distinct winglike structure, and the fixed points are now destroyed.

Therefore, it can be deduced that introducing noise to the chaotic Lorenz system takes the PDFs away from exhibiting its inherent chaos and showing distinct fixed points to approaching a unimodal and symmetric distribution, which are signs of a Gaussian distribution.

## 6 Detrended Fluctuation Analysis

### 6.1 Introduction with Langevin

The PSD of a stationary random process is the Fourier transform of its autocorrelation function (10). The autocorrelation function takes the form:

$$C(n) \approx n^{-\beta} \quad (6.1)$$

where the exponent  $\beta = 1 - \alpha$  where  $\alpha$  is the spectral exponent from PSD.  $\beta$  describes the rate of decay of autocorrelation with time.

Since the Lorenz system does not exhibit constant statistical properties with time, the traditional methods are not applicable. An alternate method referred to as the Detrended Fluctuation Analysis(DFA) does not have this requirement and generates a Hurst exponent which exhibits a robust relationship with the spectral exponent  $\alpha$  (9) which allows a detailed analysis. Equation 6.2 describes this relationship:

$$H = \frac{1 + \alpha}{2} \quad (6.2)$$

DFA is a scaling analysis that enables the detection of long-range correlations in a system (7). This method is explained below, step by step, as it is implemented on the Langevin equation.

Step 1 in DFA is to divide the cumulative sum of the Langevin equation into disjoint boxes of a range of lengths, and the local trend is evaluated for each of these box sizes for all boxes. Figure 14 shows the division of one of these box lengths and a visual representation of the local trend calculated in each disjoint box. The orange trendlines are now subtracted from the blue cumulative-sum-lines to calculate the detrended walk.

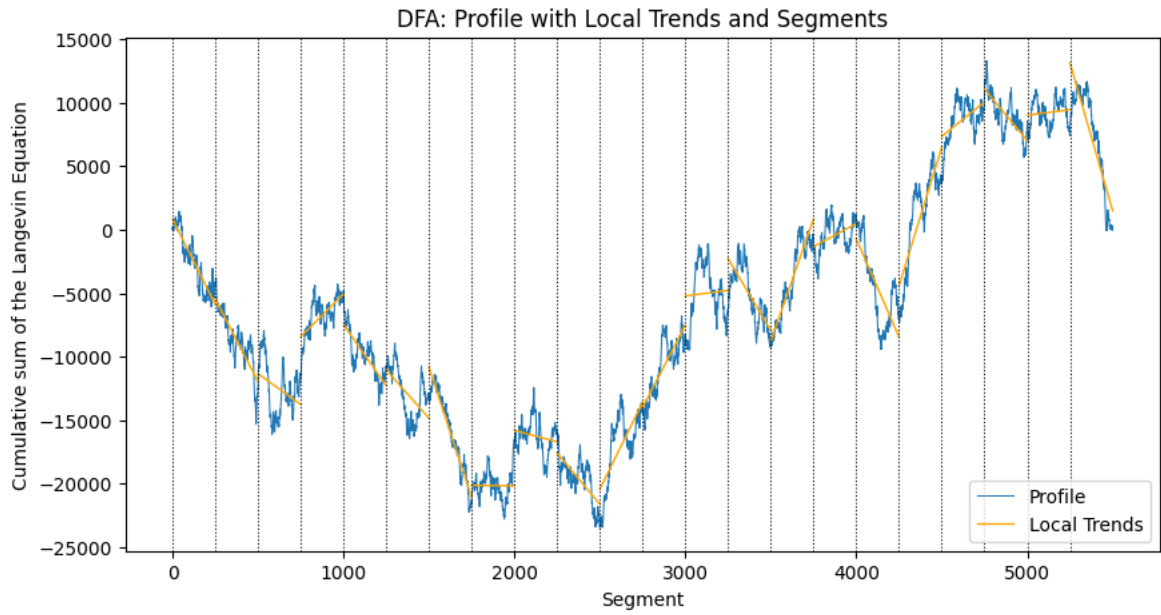


Figure 14: Local Trends

Step 2 calculates the variance of the detrended walk for each box, and the average of these variances for box size  $l$ . This is a function of  $l$ , and is now plot in a log-log space. This enables the calculation of a Hurst exponent, which is the slope and describes the scaling properties of the signal. Figure 15 shows this for the Langevin equation.

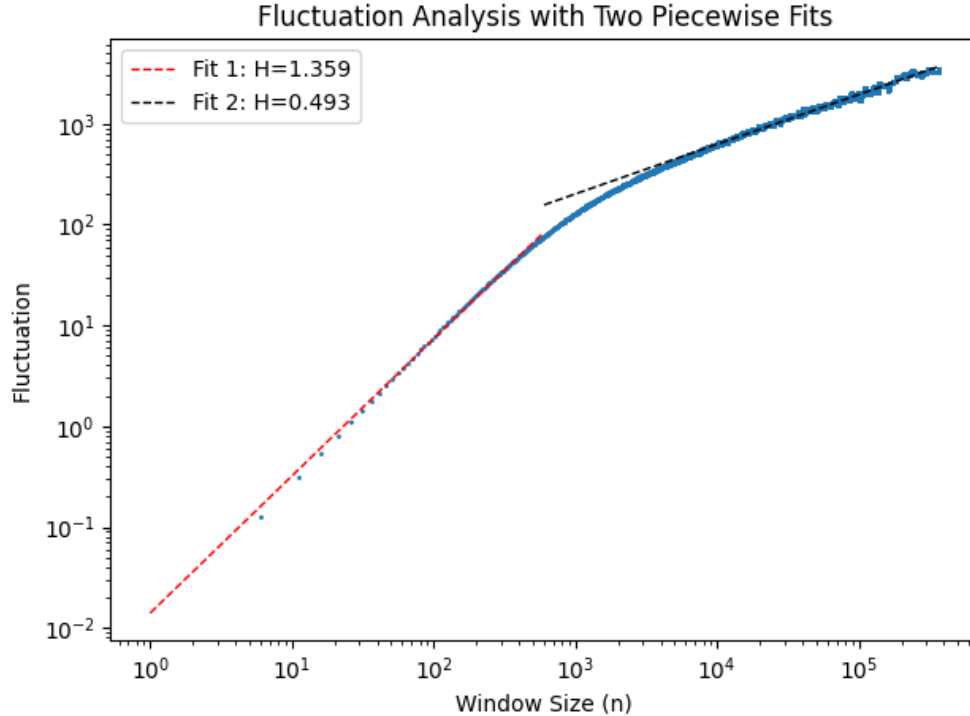


Figure 15: DFA of the Langevin equation

Step 3 would interpret this Hurst exponent to derive the correct scaling property of the model. 15 calculates two values  $H_1$  and  $H_2$  for the Hurst exponent for smaller and larger

timescales. These two values determine the local behaviour for the small scale values (smaller  $l$ ) and the global behavior for the system for the large scale values (larger  $l$ ). For the PSD, the Hurst exponent at smaller timescales,  $H_1$ , derives the spectral exponent at larger frequencies,  $\alpha_2$ .

For the Langevin equation, the DFA method closely approximates  $H_1 = 1.5$  and  $H_2 = 0.5$ , which has been established as the analytical solution (9) (12), hence validating the code used. These Hurst exponents give the corresponding values of  $\alpha_2 = 2$  and  $\alpha_1 = 0$ .  $\alpha_1$  implies white noise at longer timescales, and hence the lack of autocorrelation. However, at shorter timescales,  $\alpha_1$  indicates red or brown noise, which suggests autocorrelation.

## 6.2 DFA for the Lorenz system

The DFA method enables the derivation of the spectral constant  $\alpha$  for the Hurst exponent  $H$  for each axis  $x, y$  and  $z$  of the Lorenz system.

Figure 16 shows the profile and the DFA for the  $x$  values of the Lorenz system. For shorter timescales,  $H_1 = 1.67$  which means for higher frequencies, a PSD analysis would discover  $\alpha_2 = 2.34$  for shorter timescales, which indicates brown noise and hence suggests correlation. However, on longer timescales,  $H_2 = 0.54$  which means for lower frequencies,  $\alpha_1 = 0.08$  which signals towards approximately white noise and hence low to no correlation.

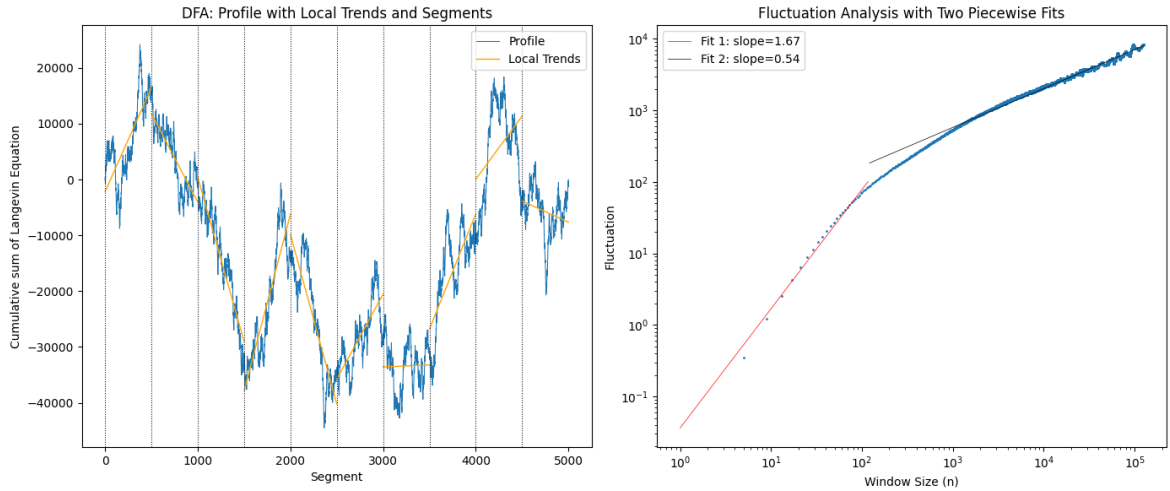


Figure 16: DFA method on  $x$  values of the Lorenz system

Figure 17 shows the profile and the DFA for the  $y$  values of the Lorenz system. For shorter timescales,  $H_1 = 1.68$  which means for higher frequencies, a PSD analysis would discover  $\alpha_2 = 2.36$  for shorter timescales, which indicates brown noise and hence suggests correlation. However, on longer timescales,  $H_2 = 0.54$  which means for lower frequencies,  $\alpha_1 = 0.08$  which signals towards approximately white noise and hence low to no correlation. These results are almost symmetrically to those observed in Figure 16, which follows intuition given the structure of the system and the similarities in the PDFs observed in Section 4.3.

Figure 18 shows the profile and the DFA for the  $z$  values of the Lorenz system. For shorter

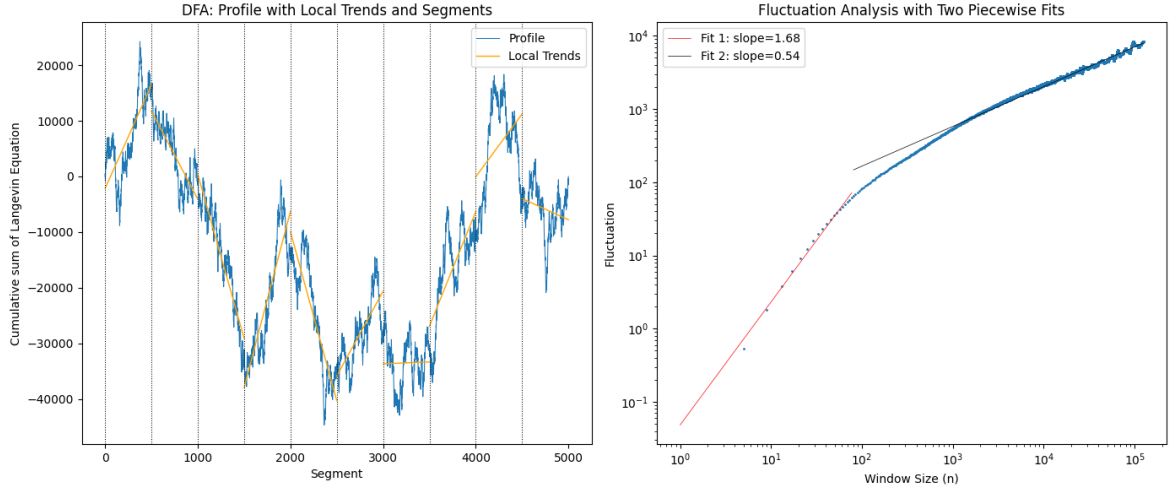


Figure 17: DFA method on  $y$  values of the Lorenz system

timescales,  $H_1 = 1.69$  which means for higher frequencies, a PSD analysis would discover  $\alpha_2 = 2.38$  for shorter timescales, which indicates brown noise and hence suggests correlation. This is consistent with the short timescale results obtained for both  $x$  and  $y$ . However, on longer timescales,  $H_2 = 0.38$  which means for lower frequencies,  $\alpha_1 = -0.24$ . While this looks problematic, it can be deciphered that  $\beta_1 = 1.24$ . When  $\beta > 1$  and  $H = 0.5$  (which  $H_2$  approximates), the process is said to have short range correlation (9).

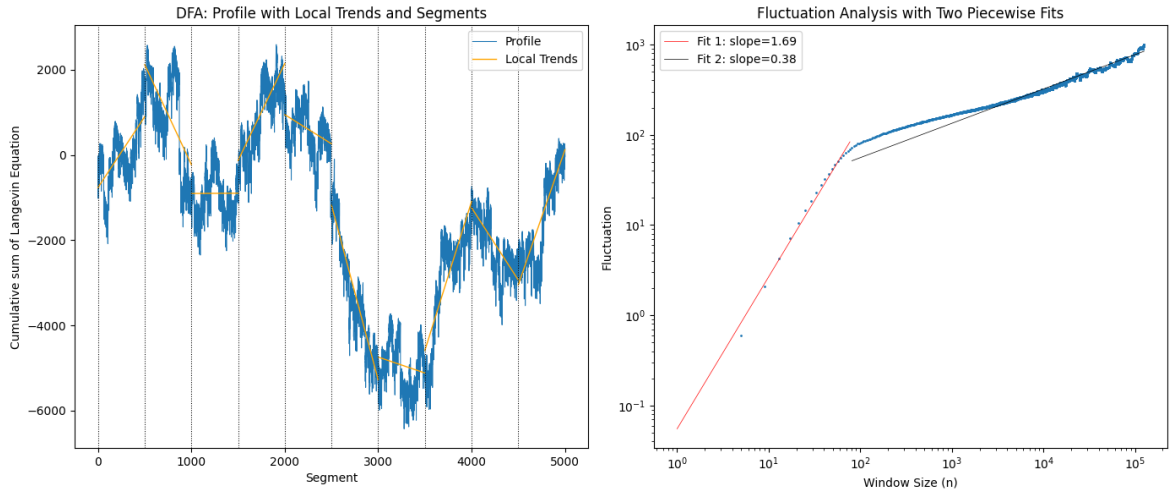


Figure 18: DFA method on  $z$  values of the Lorenz system

Figures 16, 17 and 18 establish the DFA results for the standard behaviour of the Lorenz system, and this enables the inference of changes in behaviour when noise is introduced.

Figure 19 shows the changes in the DFA along the  $x$  axis as noise is introduced into the system. At smaller timesteps, as noise increases,  $H_1$  decreases, and this causes the  $\alpha_2$  values to decrease too. These  $\alpha_2$  values still remain sufficiently close to  $\alpha = 2$ , which suggests brownian noise and some correlation. Parallely, at longer timescales too,  $H_2$  values decrease as noise increases, which results into  $\alpha_1$  values falling too. With the interpolation of noise,  $\alpha_1$  begins to take negative values, which indicates  $\beta > 1$ , and this implies short range correla-

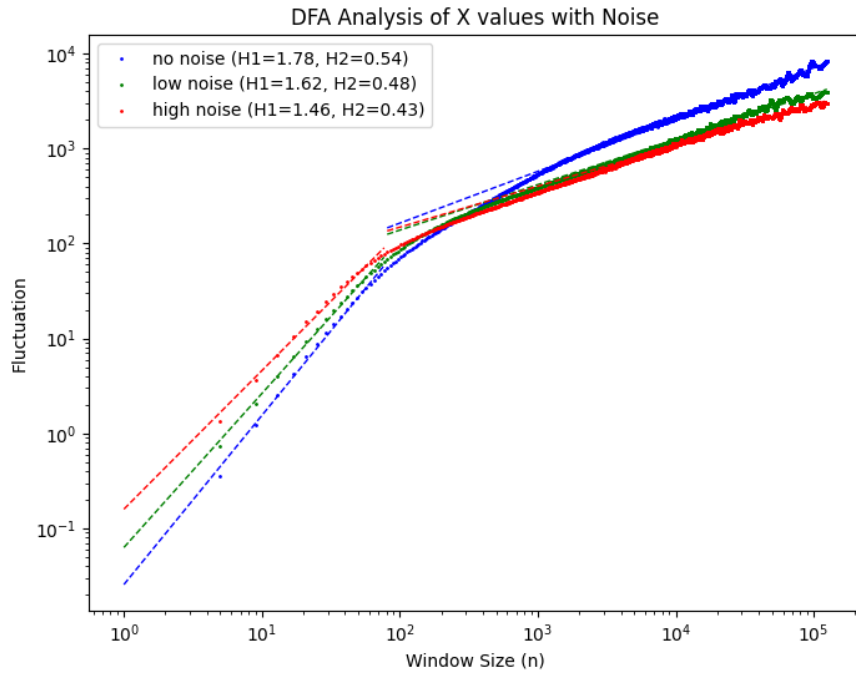


Figure 19: DFA method on  $x$  values of the Lorenz system

tion if complimented by  $H_2 = 0.5$ , which is approximately true.

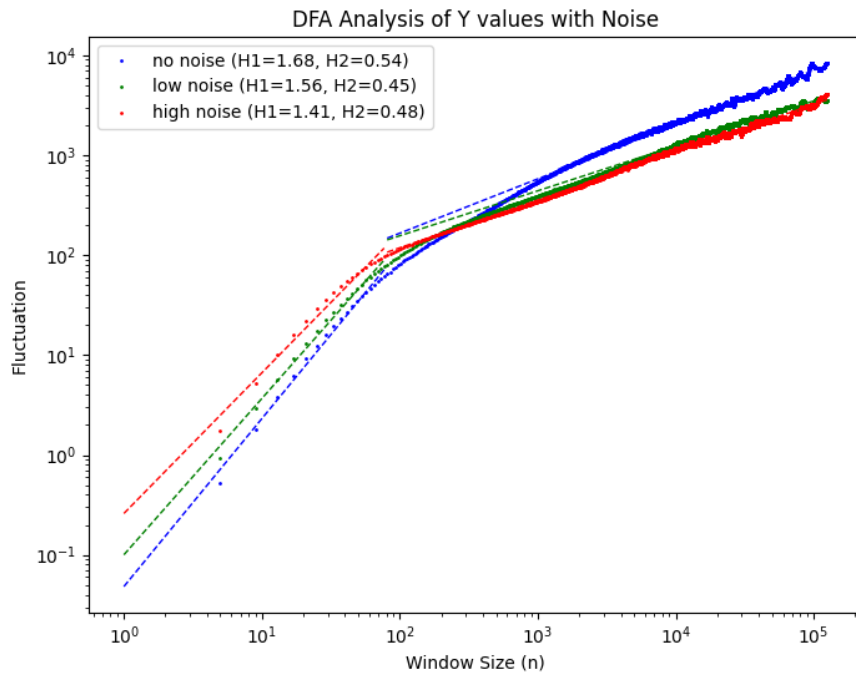


Figure 20: DFA method on  $y$  values of the Lorenz system



Figure 20 shows the changes in the DFA along the  $y$  axis as noise is introduced into the system. At smaller timesteps, as noise increases,  $H_1$  decreases, and this causes the  $\alpha_2$  values to decrease too. These  $\alpha_2$  values still remain sufficiently close to  $\alpha = 2$ , which suggests brownian noise and some correlation. At longer timescales, Figure 20 reveals a decrease and slight increase in  $H_2$  values. This is inconsistent with expectation, but can very much be attributed to the heavy truncation of the Lorenz system and the large global error. Moreover, these values still remain sufficiently close to 0.5. This reflects onto the  $\alpha_1$  values as they decrease and observe negative values as noise is introduced. This causes  $\beta < 1$ , which concludes short range correlation(9) with the addition of noise.

Figure 21 shows the changes in the DFA along the  $z$  axis as noise is introduced into the system. At smaller timesteps, as noise increases,  $H_1$  decreases, and this causes the  $\alpha_2$  values to decrease too. These  $\alpha_2$  values are arbitrarily close to  $\alpha = 2$ , which suggests brownian noise and some correlation. At longer timescales, Figure 21 reveals an increase in  $H_2$  values, which has not been observed on either the  $x$  or  $y$  axis. These values seem to be approaching  $H = 0.5$ , which means  $\alpha$  values approach 0 and  $\beta$  approaches 1 with increased noise. Since the Hurst exponent is in the range  $0 < H < 0.5$ , the process is said to have long range anti-correlation (9).

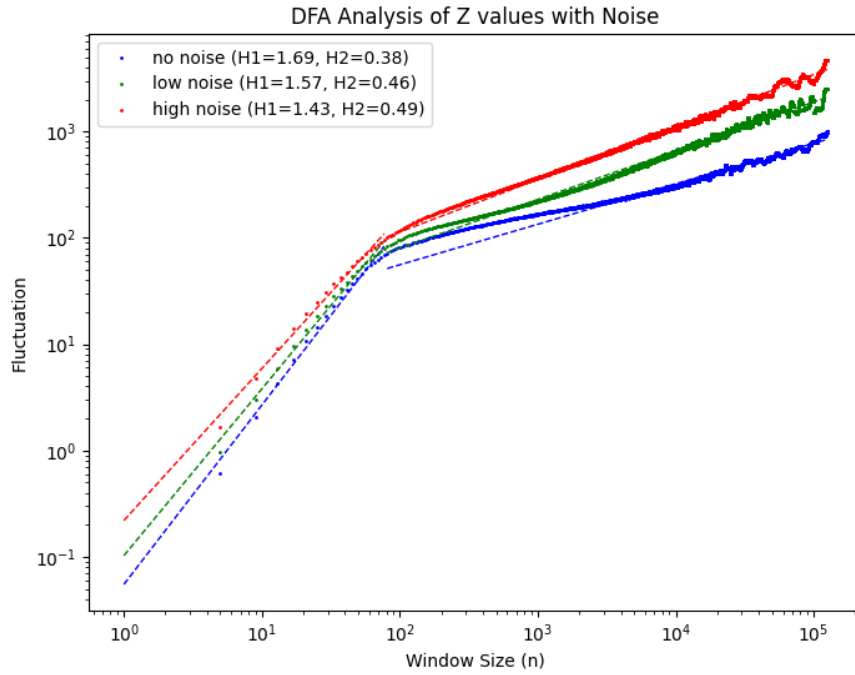


Figure 21: DFA method on  $z$  values of the Lorenz system

From Figures 19, 20 and 21 it is inferred that introducing noise reduces the autocorrelation along these axes, and hence on the overall system. This is consistent with the deduction in Section 5.4. These are the results for the Lorenz system with white noise, and noise in other colours impacts the system differently.

## 7 Conclusion

This project separates the chaos of the Lorenz system, which is governed by its nonlinearity and high sensitivity to initial conditions, to the noise added to it in the form of random normally distributed values. Since the 3 equations of the Lorenz system mentioned in Equation 2.1 are so deeply intertwined, adding noise on any one of the axes also affects the system deeply, albeit a bit slower than if it were to be introduced on all 3 axes, which is what is done throughout this project. The methods of this paper are validated throughout via implementation on the more fundamental Langevin system. Comparing the results obtained over this equation with previously conducted research enables this.

Constructing the PDFs of the Lorenz system in Section 5.4 has given insights into the standard bimodal structures on the  $x$  and  $y$  axes that are a result of the stable fixed points of the system. The PDF on the  $z$  axis is found to be unimodal. These are indicators of the determinism of the system. The introduction of noise however, results into the loss of these fixed points and all three axes now take a normally distributed unimodal PDF. This signals the loss of ordered chaos of the system, and the adoption of random behaviour. It is worth noting that this behaviour is not completely random, meaning that the attractor still accounts for its initial conditions and thus its position in the phase space, but its distinct characteristics are lost.

Section 6.2 implements the DFA method on the standard Lorenz system to infer statistical information about autocorrelation in the system over time. At long ranges, the Lorenz system without noise observes little to no autocorrelation with time. Since white noise is added to the Lorenz system, the Power Law decays faster and the system becomes increasingly random and autocorrelation disappears. This can be inferred for only white noise, and the system may behave differently under the influence of coloured noise.

It is important to address the limitations associated with this paper. The Lorenz system is a continuous process and has been discretised to enable the aforementioned analysis. This creates local errors at every step in the initial dataset, and this accumulates to introduce slight inaccuracies in the dataset. The Lorenz system is heavily truncated and the above mathematical analysis may not be put to physical use to interpret heat convection in the atmosphere, which is what the Lorenz system models. With increased precision and much larger datasets, it may be possible to use these methods for physical inference, but the scope of this project is purely mathematical.

The methods introduced in this paper can have applications in climate models and financial markets, where chaos and stochastic noise often interact. The PSD inference through the DFA method can enable better forecasting in these areas without the requirement of the process being stationary. Biological applications and fluid dynamics see an abundant use of power law statistics and these methods could be used as an introduction to further analysis in the area. Since the PSD analysis contributes towards long term forecasting, introducing stochastic noise can add a layer that can be approximated to be random which accounts for the unpredictability of the real world conditions. This makes forecasting models more realistic.

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## A Appendix

The Euler Maruyama method is an extension of the Euler method used to solve Stochastic Differential Equations (SDEs). A general SDE takes the form:

$$\frac{dX(t)}{dt} = f(t)X(t) + h(t)X(t)\xi(t) \quad (\text{A.1})$$

which develops from

$$\frac{dX(t)}{dt} = X(t) \cdot [f(t) + h(t)\xi(t)] \quad (\text{A.2})$$

Where:

$[f(t) + h(t)\xi(t)]$  is a stochastic parameter

$\xi(t)$  represents white noise

Equation A.1 can be rewritten as:

$$dX(t) = f(t)X(t)dt + h(t)X(t)dW(t) \quad (\text{A.3})$$

Where:

$dW(t) = \xi(t)dt$  denotes a Wiener process (3)  $f(t)X(t)dt$  is the deterministic part

$h(t)X(t)dW(t)$  is the stochastic part

For the simplicity of further calculations, Equation A.3 can be rewritten as:

$$dX(t) = f(X(t))dt + h(X(t))dW(t) \quad (\text{A.4})$$

For the initial conditions  $X(0) = X_0$ , the Euler-Maruyama Method can be implemented by first discretising the SDE over small timesteps of  $\Delta t = \frac{T}{L}$  over the interval  $[0, T]$ . This results into the following recursive relationship:

$$X(j\Delta t) = X((j-1)\Delta t) + f(X((j-1)\Delta t))\Delta t + h(X((j-1)\Delta t))(W(j\Delta t) - W((j-1)\Delta t)) \quad (\text{A.5})$$

for  $j = 1, 2, \dots, L$

Equation A.5 is expressed in simpler terms as:

$$X_j = X_{j-1} + f(X_{j-1})\Delta t + h(X_{j-1})(W(\tau_j) - W(\tau_{j-1})) \quad (\text{A.6})$$

Where:

$$X_j = X(j\Delta t)$$

$$\tau_j = j\Delta t$$

This method approximated the numerical solution to the SDE by discretising it, and the stochastic term introduces time step independant randomness at each step. (4)

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